

AAPL — Asian Call Option · Overnight Pricing Report

Floating-Strike · European-Style · Binomial Tree (Cox-Ross-Rubinstein, CRR)

OPTION PRICE

\$13.8805

Binomial Tree

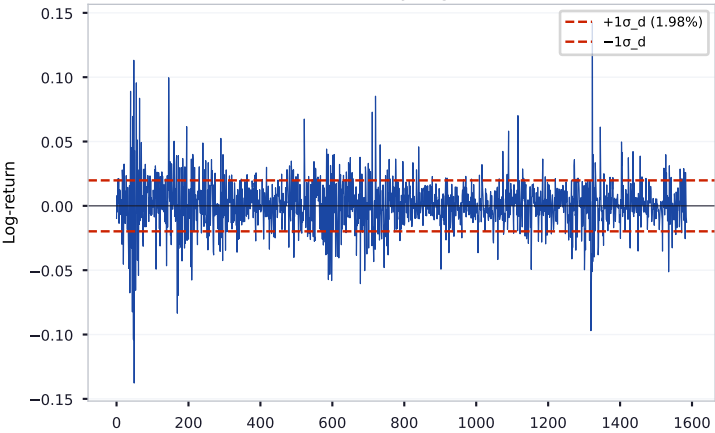
\$3.8540 · Normal Approximation

946,516 terminal states

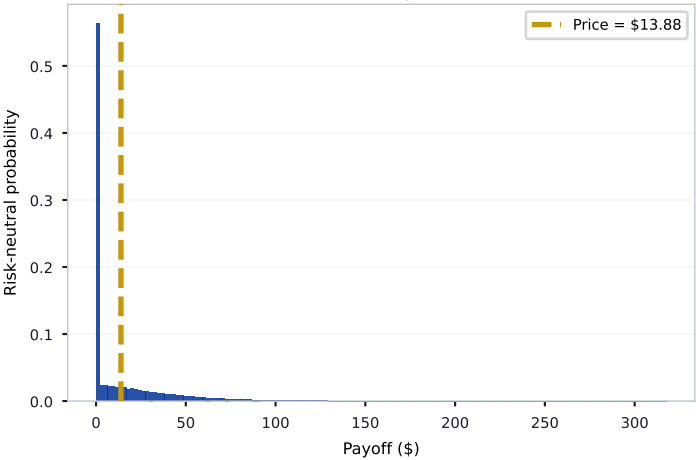
MODEL PARAMETERS

Parameter	Value	Source
Initial price	\$267.36	Dataset
Maturity	0.5 years	Input
Periods	25	Input
Risk-free (annual)	1.00% p.a.	Input
Risk-free (daily)	0.0040% / day	Derived
Period length Δt	0.0200 yrs	Derived
Daily volatility	1.980%	Est. [full]
Annual volatility	31.31%	Est. [full]
Up factor u	1.0453	Derived
Down factor d	0.9567	Derived
RN probability q	0.4912	Derived
Physical prob. p	0.5000	Assigned

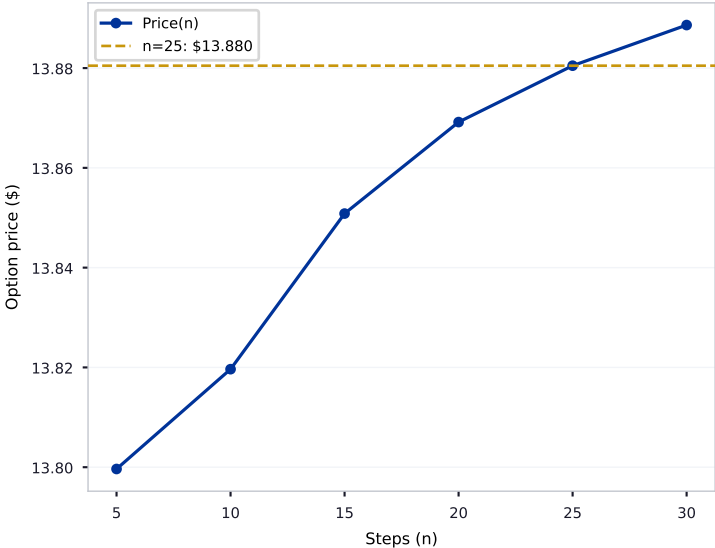
AAPL — Daily Log>Returns



AAPL — Terminal Payoff Distribution



AAPL — Step Convergence



METHODOLOGY

- Instrument

European floating-strike Asian call. Payoff at maturity T:
 $\Pi = \max(S_n - \bar{A}, 0)$
 $\bar{A}$ = arithmetic mean of all S_t along the price path
- Pricing — CRR Binomial Tree

$u = \exp(\sigma \sqrt{\Delta t})$ $d = 1/u$ $\Delta t = T/n$
 $q = (e^{r\Delta t} - d) / (u - d)$ [risk-neutral prob.]
Price = $e^{-rT} \times E^Q[\Pi]$
Path merging: states with equal (price, cum_sum) are collapsed, reducing complexity significantly.
- Volatility assumption

Backward-looking: daily log-returns computed over the full historical window (2020–present), std with ddof=1, annualised $\times \sqrt{250}$ (M=250 trading days/year, assignment spec).
 $\sigma_{\text{daily}} = 1.980\% \rightarrow \sigma_{\text{annual}} = 31.31\%$
- Physical vs risk-neutral probability

$p = 0.5$ (assigned, real-world up-move probability).
Not used in pricing — included for reference only.
Pricing is performed exclusively under Q (risk-neutral).
- Risk-free rate

$r = 1.00\%$ p.a. (annual continuous, input).
Per-step growth factor: $e^{r\Delta t} = 1.000200$